

Post-Quantum Cryptography Migration

Part 7 - Protocols & Trust: PKI, Certificates, Identity & Authentication

Compiled by the Spaced Research Ledger

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Everything that proves identity on the internet runs on certificates and signatures, and none of it is quantum-safe yet. Encryption moved first because recorded traffic is at risk today. Signatures can wait until a large quantum computer exists, but the systems that issue, distribute, and validate them take a decade to change. This is the migration's long pole.

This report covers nine layers of that problem. Section 1 covers the public certificate authorities, Section 2 the enterprise CA platforms, and Section 3 the enrollment protocols between them. Section 4 covers certificate formats and hybrids, Section 5 the enterprise authentication protocols, and Section 6 smart cards and passkeys. Section 7 covers the identity providers, Section 8 revocation, and Section 9 document signing and long-term validation.

The defining fact is stark: as of mid-2026, zero post-quantum roots chain into any major browser trust store. The defining decision is just as clear. Google, Let's Encrypt, and Cloudflare have all committed to Merkle Tree Certificates, a redesign that shrinks quantum-safe authentication data twenty-fold, rather than stuffing large ML-DSA signatures into today's certificates.

Behind the public web, the enterprise stack is further along. Microsoft's certificate services, AWS's private CA, and the major open-source CA platforms all issue ML-DSA certificates today. The identity layer trails everything, with no standard yet for post-quantum SAML, and passkeys still signing with algorithms a quantum computer would break.

1. Public CA Ecosystem

The public certificate authorities, governed by the CA/Browser Forum and the browser root programs, decide what certificate the padlock trusts. Change here requires coordinated rules, audited CAs, and browser support, which is why it moves slower than any software release. This section covers an ecosystem that has chosen its path but not yet taken it.

Recent Developments

The industry picked an architecture instead of an algorithm swap. The IETF chartered the PLANTS working group to standardize Merkle Tree Certificates, with Google, Cloudflare, Let's Encrypt, and independent cryptographers at the table [1]. Chrome declared MTCs its preferred path, and Cloudflare is running a live feasibility experiment against real Chrome traffic [1]. The efficiency argument is dramatic: an inclusion proof covering millions of certificates is 736 bytes, with no per-certificate post-quantum signature on the wire [1].

Let's Encrypt committed to the same path in June, targeting staging issuance in late 2026 and production in 2027 [2]. It noted the rollout touches its issuance systems, the ACME protocol, revocation, and logging [1]. The first CA application for Chrome's MTC test program arrived in August [3], and the draft keeps iterating

[4]. DigiCert published a developer playground [1] alongside recommendations favoring plain ML-DSA over composites for public trust [5].

Meanwhile the rulebook has not moved. No CA/Browser Forum ballot enabling publicly trusted post-quantum TLS certificates had been adopted by late August 2026 [6], consistent with earlier tracking [7]. The count of ML-DSA roots in the Mozilla, Apple, Microsoft, or Chrome stores remains exactly zero [8].

Current State

The reason authentication lags encryption is size. ML-DSA-44 signatures run about 2,420 bytes against 64 for classical ECDSA. A handshake carrying five signatures could push past a threshold estimated to break several percent of real connections [9]. Chrome reaffirmed in August that it has no plan to add traditional post-quantum X.509 certificates to its root store, pursuing MTCs with no published timelines [10].

The working group's draft defines the new certificate form with public logging built in, cutting overhead for short-lived certificates and large algorithms alike [11]. Under the MTC design, transparency becomes structural: a certificate cannot exist outside the published tree, and the proof is itself proof of publication [9]. Let's Encrypt describes today's transparency ecosystem as bolted on after the fact, with separate logging signatures riding in every handshake [12]. Tree heads will be signed with hybrid classical-plus-quantum keys [13], and log entries store hashes rather than full keys for a hundred-fold saving [13].

Let's Encrypt already runs the same data structure in its transparency logs [14]. Google proposed a dedicated quantum-resistant root store accepting only MTCs [13]. Cloudflare's experiment re-encodes about a thousand real certificates with X.509 fallback [15], and the company frames key exchange as half done and certificates as the harder half [16][17]. The earlier announcement dates from October 2025 [18].

What the Forum has passed is preparation. Ballot SMC013 enabled pure post-quantum certificates in the S/MIME requirements for experimentation [19][20], with root programs free to add their own conditions [19]. Ballot SC-081v3 shrinks maximum TLS certificate lifetime in stages to 47 days by 2029 [21][22], and a separate ballot retired SHA-1 from S/MIME [23]. A TLS-specific post-quantum ballot was still in drafting as of late 2025 [17], with industry expectations pointing to late 2026 or 2027 [7].

The Forum is also removing client-authentication usage from public certificates, pushing those uses to private PKI [24]. Its April meeting heard research suggesting ECC may fall to quantum attack more cheaply than thought, without any ballot proposed [25]. It also heard progress reports on MTCs, including open-source implementation work [25]. Commentary connects the lifetime shortening to algorithm agility [26].

At the CA product level, post-quantum today means private, not public. DigiCert issues private ML-DSA and SLH-DSA certificates [27] but not public ones [28]. It excludes Falcon and composites as experimental [27], and supports CRL but not yet OCSP checking for the new algorithms [29]. It added hardware-backed issuance through a validated HSM [27]. Sectigo runs a sandbox for post-quantum certificate experimentation [30] and launched managed private post-quantum issuance in April [31], covering all three ML-DSA sets [32], hosted with a virtual HSM [31].

Europe's regulatory layer is converging on the same timeline. eIDAS 2.0's technology-neutral language lets the EU Digital Identity Wallet transition later without mandating post-quantum now [33], and no binding European technical standard for post-quantum qualified certificates exists [34]. The trust-service chain still

runs entirely on classical cryptography [35][36], with ETSI's migration framework prioritizing the store-now-decrypt-later exposures [37][37]. Qualified website certificates remain classical [38].

The EU's coordinated roadmap wants national transition strategies started by the end of 2026 and critical infrastructure done by 2030 [39]. A proposed directive would write post-quantum policy into law, naming the wallet ecosystem's certificate-heavy PKI as one of the longest-lead migrations [40][40][40]. The wallet rollout itself proceeds classically, with the December 2026 deadline unchanged [41]. The validation-queue reality bites here too: with a documented backlog and few fully validated hardware modules, qualified-signature regimes cannot yet deploy post-quantum compliantly at scale [42].

Unresolved Conflicts

Two tensions stand. Google's February plan described a concrete second phase inviting log operators in early 2027 [15], while Chrome's August policy page says there are no planned MTC implementation timelines [10]. Whether the timeline was walked back or merely omitted is unclear, and a dated Chrome roadmap would settle it. Separately, whether the 47-day certificate ballot was quantum-motivated is disputed: its own text cites revocation reliability, while commentary retroactively frames it as post-quantum groundwork [26].

2. Enterprise CA Platforms

Inside organizations, certificate authorities issue the credentials that machines and employees use every day. These platforms answer to no browser root program, so they can move as fast as their vendors ship. This section covers the layer where post-quantum certificates stopped being theoretical, and where the gaps between products are now wide.

Recent Developments

Microsoft's certificate service crossed into general availability. ML-DSA shipped in Active Directory Certificate Services in May 2026, the first post-quantum algorithm there, with no in-place migration: a new parallel hierarchy is required [43]. The documentation lays out the phases, with ML-KEM and composites planned next [44]. Among the others, EJBCA now covers ML-DSA, ML-KEM, composites, and SLH-DSA with HSM interoperability [45].

HashiCorp Vault's signing engine added SLH-DSA and hybrids [46], exposed through its infrastructure tooling [47]. CyberArk's certificate manager documented new composite algorithms [48] and made post-quantum available by default, invalidating old experimental keys on upgrade [49].

Current State

The Microsoft stack is furthest into production. AD CS supports ML-DSA for CAs, templates, and responders [50], phased with ML-KEM later [51], covering all three parameter sets across the full lifecycle [52][53]. General availability landed in May [54], following the platform APIs in November 2025 [55], enabling pure ML-DSA certificates for code signing, TLS, and OCSP [44]. It is confined to Windows Server 2025, with no backport plans [56], and HSM vendors are already publishing integration notes [57].

AWS made its private CA post-quantum in November 2025. It creates ML-DSA CAs and certificates with revocation support across all regions [58][58] and combines with its key service for code-signing hierarchies [59][60]. The service builds on the first FIPS-validated open-source module with ML-KEM [61], and cites the federal 2030 and 2035 deprecation timeline [62].

The open-source and commercial CA field is a capability patchwork. EJBCA replaced the draft algorithms with the finalized ones in 9.1 [63][64], though its key-recovery keys stay RSA and OCSP signing remains classical [64]. It solved the KEM enrollment problem, since KEM keys cannot sign requests, through encrypted-certificate proof of possession [65] and an alternative message format [66]. It offers dual-key Chimera chains [67], composite keys [66], experimental Falcon [68], and hash-based schemes per the industry capability matrix [69], with newer releases broadening support [70][71].

Its vendor marks experimental algorithms as non-production [72], expanded across its signing products [69] [70], launched a test-drive lab [73], and bought two cryptographic-discovery companies [74]. Keyfactor's documentation aligns with the ratified hardware-token standard [65] and splits community from enterprise support [67], with its stated commitment to adopt finalized standards [64].

Entrust sells hybrid and pure issuance backed by validated HSMs [75][76] and helped define the IETF hybrid formats [77]. It offers readiness assessments [77], and published survey data showing organizations broadly unready [78]. Its own HSM matrix still shows the hash-based schemes missing [69][70].

The stragglers are instructive. Vault's post-quantum support lives in its signing engine, explicitly for evaluation [79][80], with the PKI engine's request still an open issue [81][82] and hybrid plans waiting on upstream maturity [83][84][85]. smallstep implemented hybrid TLS but stalled on certificates, judging ML-DSA too large and slow pending NIST's follow-on competition [86], with community discussion noting no client demand yet [87]. Its CA remains classical-first [88], supporting the hybrid key exchange off by default [86], with the matrix showing ML-KEM but no signatures [69].

Venafi added the algorithms across two releases [89][90], then changed owners twice, landing inside Palo Alto Networks [91]. A rival's claim that the new owner deprioritizes PKI is competitor marketing, not verified fact [92].

3. Certificate Enrollment Protocols

Certificates reach devices through enrollment protocols: ACME for web servers, SCEP for phones and network gear, EST and CMP for industry and IoT. Most were designed assuming a key can sign its own request, which post-quantum KEM keys cannot do. This section covers how each protocol is adapting, or failing to.

Recent Developments

ACME is growing post-quantum profiles. New drafts define quantum-ready profiles [93] and per-account algorithm preferences [94]. The important one solves proof of possession: a public-key challenge letting a client prove it holds a private key without a self-signed request, built for KEM keys [95]. It is now in a formal adoption call [96]. SCEP, by contrast, has no ratified post-quantum extension at all [97].

Current State

ACME's position is mostly inherit-and-wait. The protocol standardized in 2019 [98] automates validation and now underlies most public certificates [99], with extensions keeping it current as lifetimes shrink [99]. It contains no algorithm negotiation [100], a gap one expired draft tried to fill [100][101]. The IETF's status page lists it as needing no protocol-specific work since it inherits from its building blocks [101]. Enterprises increasingly use it beyond the web, with SCEP kept for older devices [102].

CMP did the real KEM work. The protocol's 2023 refresh improved algorithm agility [103], with a lightweight profile for industrial use [104] and a maintained algorithm registry [105]. The 2025 revision, RFC 9810, adds explicit management of KEM-key certificates [106], the culmination of a multi-year design arc [107]. A shared CMS building block carries KEM support across protocols [101], and the composite-signature draft serves them all while remaining unratified [108], with its keys restricted to signing-only use [109].

EST and SCEP sit at opposite ends of dignity. EST profiles enrollment over TLS [110], with mutual authentication and simple re-enrollment [110], errata fixes [111], a template mechanism for request contents [112], and a renewal-guidance draft [113]. The tracker lists it as inheriting post-quantum support [101]. Microsoft's CA does not speak it natively, requiring gateways [114].

SCEP was standardized informationally in 2020 after twenty years [97] and carries requests in CMS envelopes [97]. It gained non-RSA enrollment late [115], and rests on a preshared-password model weaker than EST's [116]. It remains the most deployed protocol for device management [116], implemented even in new products [117]. Research proposes direct post-quantum modifications [118], while CA vendors openly question whether it survives the transition [119].

The tracker itself is unsure which working group owns it [101].

4. Certificate Formats & Hybrid Constructions

A hybrid certificate carries both a classical and a post-quantum key, insurance against either one breaking. The IETF's composite drafts weld the two into one atomic algorithm, while older dual-key designs let legacy systems ignore what they do not understand. This section covers the formats, and the timestamping machinery that must outlive them all.

Recent Developments

The composite specifications are inches from the finish. The composite KEM draft entered IETF Last Call in July [120], advanced to IESG evaluation with a September telechat [120], and its version history shows steady progress [121][122]. On the TLS side, drafts define composite ML-DSA negotiation [123] and a continuity extension letting clients pin a server's post-quantum commitment against downgrade [124]. A separate proposal would bind certificates to a committed future algorithm by hash [125].

Current State

The terminology and rules are now formal. RFC 9794 defines the post-quantum/traditional hybrid taxonomy, with composite as the instantiation that looks like a single algorithm to unaware systems [126][127]. Some

national authorities already mandate lattice schemes only in hybrid form [128]. The composite signature draft cleared Last Call in January [129] yet kept revising through April without becoming an RFC [130][108], defining fixed pairings of ML-DSA with the classical algorithms [127].

It is positioned as extra insurance rather than a universal replacement [131], with signing-only key usage enforced [132]. Companion drafts carry composites through CMS [133][134]. A hardening proposal binds signatures to the composite key to block cross-key forgery [135], and the KEM variant's design goal is drop-in compatibility across the certificate ecosystem [126]. A FIPS-certification rationale drives the KEM composite: pair an uncertified ML-KEM with a certified classical component and the module can certify now [136].

An IETF hackathon project maintains public interoperability matrices [137]. Vendor guidance still calls composites not production-ready, recommending plain post-quantum certificates as the simpler path [138].

The dual-key alternative took a different road. The original catalyst draft is formally dead at the IETF [139]. Its approach lives on in the ITU-T X.509 standard, where a certificate carries an alternative key that legacy parties ignore [139]. Practitioners now distinguish three families: composites, catalyst-style alternatives, and chameleon dual certificates [140], with several products supporting catalyst certificates while chameleon remains a draft [69].

An academic comparison notes the catalyst draft expired without becoming the international mainstream [141]. A separate individual draft proposes certificates signaling continuity commitments [142], carrying no formal standing [143]. One structural rule outlasts every format choice: a chain is only as post-quantum as every signature in it, which is why PKI is the long-lead item [144].

Timestamping is quietly ready. RFC 3161's structures are not algorithm-specific, so migration is substitution, not rewrite [145][146], with long-term evidence preserved through existing renewal mechanisms [146]. Commercial servers already sign tokens with ML-DSA [147], one vendor cross-validated ML-DSA timestamp tokens against independent tooling [148][149], implementing the ratified algorithm-identifier RFCs [148]. No standards-track update to the protocol itself exists or is needed [150].

The tokens are ordinary X.509 signatures facing the same migration [151]. Analysts warn timestamps are only partial mitigation, and that revocation endpoints migrate separately [152][144]. A new RFC carries the tokens into COSE structures [153].

5. Enterprise Auth Protocols

When an employee signs into anything, the login flows through SAML, OpenID Connect, Kerberos, or LDAP, all of them signing and encrypting with classical algorithms. The tokens are short-lived, but the signing keys and the directory traffic are not. This section covers the protocols where post-quantum work has barely begun, and why some will migrate far more easily than others.

Recent Developments

The token-format layer got its standard. RFC 9964 registers ML-DSA identifiers and a new key type for JOSE and COSE [154], the piece every JSON-based login protocol was waiting for. The registrations are formal IANA entries, all marked recommended [155]. The working group continues on two tracks, with draft identifiers

subject to change until registration completes [156], a composite-signature serialization draft [157], and the KEM draft rescope to COSE [158].

Size math stays manageable for web tokens carried over HTTPS [156]. Kerberos moved too: an individual draft extends its certificate-based pre-authentication with ML-KEM [159] and reached a first revision [160]. It was interop-tested at an IETF hackathon with a working MIT Kerberos build [161]. Microsoft's directory documentation reiterates the phased certificate rollout [44][162], while a vendor whitepaper confirms the platform's hybrid TLS covers LDAPS without any directory-specific mechanism [163].

Current State

The JOSE and COSE plumbing is the furthest along. The ML-DSA serialization draft defines the new key-pair type [164][164], and hybrid public-key encryption drafts cover the KEM side [165]. The original multi-algorithm draft was split per scheme [166], with the per-algorithm drafts active except an expired Falcon one [101]. Hash-based signatures have had a COSE RFC for years [101].

Kerberos has a design but no deployment. The draft defines its KEM structures, key derivation, and downgrade prevention while remaining individual, not adopted [159][167]. Microsoft lists Kerberos on its roadmap with progress expected into 2027 [168]. Red Hat states plainly that nothing beyond TLS and SSH supports post-quantum signatures in its current release, Kerberos included [169].

Directory traffic is the quiet exposure: advisories flag LDAP, Kerberos, and DNS between endpoints and domain controllers as harvest-now-decrypt-later targets [170][171]. Microsoft's certificate support explicitly does not extend to the directory-authentication layer itself [172], though issued certificates cover domain authentication uses [51][51][44], with composites on the forward roadmap [168].

The federation protocols split cleanly. OIDC signs tokens with quantum-vulnerable algorithms like everything else [156]. But its identifiers live in IANA registries where drafts exist, and its keys distribute through polled endpoints, making its path structurally cleaner [156]. The real near-term advice is defensive: tokens are short-lived, so protect the long-lived signing key and the endpoints instead [156].

Government profiles already demand crypto-agility and recommend the hybrid TLS suites [173][173][173]. A research prototype shows a full post-quantum OIDC stack works [174], while no RFC defines post-quantum token signing yet [175].

SAML is stuck at a lower layer. No specification exists for post-quantum SAML signatures [176], and its XML signature format has no post-quantum algorithm URI to reference [156][156]. Migration would mean coordinated metadata redistribution across every service provider [156]. None of the major providers documents general-availability post-quantum SAML signing [156].

Guidance points to ML-DSA as the eventual replacement, once standards exist [175][175]. The deeper problem is organizational: SAML's standards committee no longer exists, so the work would have to happen in the XML-signature world [177].

6. Passwordless & Hardware Tokens

Smart cards and passkeys are the physical end of identity: the badge that opens the federal building, the security key that defeats phishing. Both sign with elliptic curves a quantum computer would break, and both live in hardware that cannot be patched. This section covers the standards bodies designing their successors, and the years-long gap before anything ships.

Recent Developments

The U.S. federal credential started its redesign. NIST released working drafts updating the PIV smart-card standards for ML-DSA and ML-KEM [178]. They rest on a dual-stack model adding post-quantum containers beside the classical ones for incremental deployment [178][179].

The drafts are explicitly labeled preliminary [178], with a gap analysis attached [180]. Research found KEM-based card authentication far lighter than signatures on constrained hardware [181], and IDEMIA demonstrated the first complete PIV application using it [182]. NIST contracted support for the credentialing update [183], and GSA is rebuilding the federal identity architecture around post-quantum, with pilots through 2028 and full migration targeted by 2035 [184].

The passkey world reached milestones without post-quantum. WebAuthn Level 3 became a full W3C Recommendation in August [185], with an open post-quantum transition issue on the next milestone [186]. Yubico demonstrated ML-DSA running on YubiKey hardware [187], and a vendor partnership is developing end-to-end post-quantum FIDO2 passkeys [188].

Current State

The FIDO stack has groundwork but no shipped product. Current implementations sign with classical curves [189][190], IANA registered the ML-DSA identifiers in 2025 as the prerequisite [191][192], and the server requirements already list ML-DSA as recommended or optional [193]. The Alliance runs a dedicated post-quantum working group, with no finalized specification or platform implementation yet [192][194]. Google's 2023 hybrid security key was a research demonstration predating the registrations [191][195].

Researchers name the missing secure hardware and the absent roadmap as the bottlenecks [196]. Academic work evaluates ML-DSA's fit [197], and proposed architectures treat the migration as lifecycle management across every component [198].

The protocol versions keep the platform current. CTAP 2.3 published in February as the certification baseline [199][200], backward-compatible [199], adding transport and usability features [201], with certification consolidated on it [201]. Its predecessor added token and policy features [202], and CTAP pairs with WebAuthn to form FIDO2 [197][203]. WebAuthn Level 3 consolidates passkey behaviors and adds capability APIs [204], after a candidate phase [185][205], atop the earlier Recommendations [205].

RFC 9964's publication gave the web-token side its identifiers [206], and a draft describes ML-DSA in WebAuthn directly [207]. Adoption is the bright spot: roughly 5 billion passkeys active [208], with high awareness and enterprise deployment [208], and portability arriving through a credential-exchange protocol [208].

The federal card fleet is the hard case. Current standards authenticate with RSA or ECDSA at graduated assurance [209], the defense card remains the standard credential [210], aligned department-wide on PIV authentication [211]. Policy now admits passkeys as approved alternatives for some uses, with phishable authenticators to retire by fiscal 2027 [212]. A live GSA pilot issues PIV credentials with classical, hybrid, and post-quantum profiles [213]. The physics problem is memory: legacy chips with as little as 72 kilobytes cannot hold post-quantum material, so dual-signature certificates let old systems fall back to classical [214].

7. Identity Provider Platforms

A handful of identity providers hold the signing keys behind millions of corporate logins. When they migrate, whole ecosystems follow. Until then, every relying application waits. This section covers a market of roadmaps, where the clearest engineering commitment comes from an open-source project rather than any commercial vendor.

Recent Developments

Microsoft moved its target forward. The company now aims to transition its products and services, Entra included, to post-quantum by 2029, four years earlier than its prior 2033 goal [215]. Its security initiative reports a defined transition approach across certificate services and identity platforms [216]. Okta paired its hybrid key exchange rollout with tooling to migrate customer tenants as defaults change [217], positioning its coverage across all hyperscalers [217].

Keycloak's community opened concrete feature requests for ML-DSA token signing [218] and hybrid composite signatures targeting a named release [219]. An independent ranking credits Ping's orchestration engine as a natural phase-in path while noting 2026 readiness mostly means agility, not shipped features [220].

Current State

Microsoft's identity migration is roadmap plus foundations. The published plan makes Entra authentication, passkeys included, quantum-safe on the long timeline [221]. The underpinnings arrive earlier: certificate services issuing ML-DSA [53], and platform APIs generally available [222][55]. Hybrid TLS sits in the stack beneath the services [223], with standing involvement in the open-source and NIST integration efforts [223]. Authentication scenarios themselves, Kerberos and passwordless included, are explicitly future work into 2027 [224].

Keycloak is the transparent one. It supports only classical algorithms today [225], with a readiness epic explicitly keeping post-quantum optional and legacy algorithms intact [226]. Maintainers report the signature mechanics easy and the key-distribution standards the hard part [225], and recommend hybrids despite format gaps [225]. They field adjacent requests from password hashing [227] to a years-old dependency bump [228].

The work in progress spans SAML readiness [229], a priority research spike [230], and a sixty-subtask epic targeting a named milestone, more concrete than any commercial provider's public plan [231]. The motivating deadlines are regulatory, led by France's 2027 certification cutoff [231]. Outside observers tie the

work to the new token-signing RFC, noting that the first post-quantum token an IdP issues obligates every downstream verifier [232].

The commercial providers publish posture, not features. Okta's two-phase plan starts with hybrid key exchange against harvest-now-decrypt-later [217][217], wrapped in customer audit guidance [217][233], and its appearance in a conference promotion names no shipped capability [234]. Ping evaluates the standards and plans adoption as they mature [235], positions its platform around agility [235], and publishes migration playbooks [236]. Its executives urge inventory-first preparation [237][238] while calling the shrinking window real [239].

Unresolved Conflicts

Microsoft's own dates disagree. The current announcement targets 2029 for the product transition [215], while the standing Entra roadmap document still says full transition by 2033 [221]. The likeliest reading is acceleration that has not propagated through all documentation, but the documents themselves do not say so. A single updated Microsoft timeline would settle it.

8. Revocation & Status Infrastructure

When a certificate goes bad, revocation lists and OCSP responses are how the world finds out. Both are signed artifacts, so post-quantum signatures multiply their size at exactly the moment the industry was already losing faith in them. This section covers the arithmetic, and the two escape routes: certificates too short-lived to revoke, and Merkle trees.

Recent Developments

The measurements arrived and they are ugly. Profiled revocation lists grow roughly seven-fold under ML-DSA-44 and forty-fold under the larger hash-based sets [240]. OCSP responses grow from about 2 to 8 kilobytes, list entries from 300 bytes to 4 kilobytes, and a hundred-thousand-certificate database from 240 megabytes to 1.8 gigabytes [241][241]. The Merkle Tree Certificates draft moved onto the standards track [242], with deployment use cases [243] and bulk proof mechanisms [244].

Chrome has a phased plan running through a 2027 root-store finalization [245]. The editor's copy clarifies that classical revocation applies unmodified to MTCs [4]. Let's Encrypt's plan folds revocation tooling into its MTC transition [2]. A new draft proposes stapled status via sparse Merkle proofs and threshold signatures as an OCSP alternative [246].

Current guidance recommends hours-to-days certificate lifetimes for high-scale systems precisely to sidestep revocation [247].

Current State

The size problem compounds through the stack. ML-DSA inflates chains, lists, and responses together [248], and the hash-based schemes are worst for revocation artifacts [248]. Modeling of RPKI-style object stores shows growth from 1.2 gigabytes to 2.9 under ML-DSA and to 39 in the worst hash-based case [249][249]. A

response's penalty stacks on the chain's, since responses embed certificates [250], and a three-tier ML-DSA chain crosses 20 kilobytes before the handshake starts [250].

NIST proposed reduced-size hash-based parameter sets aimed at certificate signing [251]. Forum deliberations even floated hard byte caps forcing CAs to partition lists [252].

The industry's direction was set before the algorithms arrived. The Forum made OCSP optional and lists mandatory in 2023 [253], and Let's Encrypt completed its OCSP exit in 2025, decommissioning its responders entirely [254][255]. Where OCSP survives, stapling carries larger signed responses through an otherwise unchanged protocol [256], recommended alongside caching extensions to absorb the size [257]. Stapling already fails silently across much of production [258].

Short lifetimes are the working substitute. The 47-day schedule passed with every browser in favor and zero CAs opposed [259][260], its text framing revocation as complementary and sole reliance on it as untenable [261][262]. Bounded lifetimes bound algorithm exposure too, which is the crypto-agility argument [263]. MTCs push the same logic further, recommending hourly batches and 14-day certificates so expiry replaces revocation [264].

Within the MTC design, relying parties revoke by index ranges without knowing log contents [265], and an optional mode drops signatures entirely for current clients [266]. The log entries dodge the size penalty by storing hashes [265], and a CA's cosigner key may still sign classical status artifacts [266]. Researchers classify the whole construction as research-stage, against deployed classical schemes and emerging plain ML-DSA [267][267]. Meanwhile Microsoft's responder signs OCSP with ML-DSA today [268][269], the concrete counterpoint to all the redesigns.

9. Long-Term Signature Validation & Document Signing

A signed contract must stay verifiable for decades, long after the algorithm that signed it dies. Europe's advanced-signature standards solved this for classical cryptography with layered timestamps and archival levels. This section covers how that machinery absorbs the post-quantum transition, and the e-signature platforms dragging behind it.

Recent Developments

Europe's algorithm catalogue went post-quantum. ETSI TS 119 312 version 2.1.1, published June 2026, incorporates ML-DSA, SLH-DSA, the hash-based schemes, and hybrids, with recommended suites and a dedicated migration section [270][270]. Downstream assessors updated their policies immediately [271], and ETSI opened a new report series on post-quantum impact across signatures, trust providers, and the EU wallet [272]. Among the platforms, DocuSign started implementing ML-DSA with dates promised in 2026 [273].

Current State

The standards were built algorithm-agnostic, which is now paying off. PAdES, the PDF signature standard, only recommends the catalogue's algorithms rather than requiring them, so it can carry post-quantum

signatures without a specification change [274]. Its current version [275] defines four baseline levels [276], with the archival level designed to outlive algorithm expiry by accumulating new material [277][278]. The XML equivalent augments incrementally the same way [279][279], with archival timestamps requiring embedded validation material first [280] and profile-level placement rules [281]. The CMS-based standard sits at its 2023 revision [282][283], and a new COSE-based signature standard defers to the catalogue by construction [284].

The catalogue is also the migration clock. The superseded version already set end dates rather than open-ended approvals [285][286], and a machine-readable schema encodes them for automated preservation [287]. The post-quantum revision arrived roughly on its announced schedule [286]. The forcing function is simple: suitability periods are shorter than statutory retention, so long-lived signatures must be re-timestamped before their algorithms expire, a maintenance process the standard mandates [285][285]. Interoperability rests on the coordinated European mechanism agreements [286].

The platform gap is the story to watch. Adobe states no confirmed post-quantum roadmap for Acrobat signatures as of March 2026 [288]. Yet the first ML-DSA-signed PDF passed between a vendor and Adobe in May 2025, SLH-DSA followed, and the PDF industry agreed to formalize both algorithms into the specification [289][289]. Adobe's transport layer meanwhile keeps supporting old TLS for lagging clients, with post-quantum interest noted [290].

Europe's qualified-signature regime frames the deadline. eIDAS 2.0 requires every member state to issue a digital identity wallet by the end of 2026 [291]. Wallet-created qualified signatures carry legal equivalence to handwritten ones through qualified providers [292]. It expands the trust services [292] and sets identity-proofing requirements on issuance [293], with existing providers coexisting through the transition [294]. All of it, for now, on classical cryptography, which is exactly why the catalogue's new migration section matters.

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